Question	Answer	Marks	Partial Marks
4(a)(ii)	$\frac{9}{25}$ oe	3	**M2** for $\frac{2}{5}\times\frac{3}{5}+\frac{3}{5}\times\frac{1}{5}$ oe or **M1** for one of these products OR **M1** for probability tree identifying all 25 outcomes with the correct 9 identified OR **M1** for completed possibility space / 2-way table identifying the 9 possible outcomes out of 25, oe
4(a)(iii)	Jojo and e.g. $\frac{40}{100}>\frac{36}{100}$	1	**1FT** *their* **(i)** and **(ii)** dep on being in range 0 to 1
4(b)	$\frac{24}{60}$ oe	3	**M2** for $\frac{2}{5}\times\frac{3}{4}\times\frac{1}{3}+\frac{3}{5}\times\frac{2}{4}\times\frac{1}{3}+\frac{3}{5}\times\frac{2}{4}\times\frac{2}{3}$ oe or **M1** for any one correct product OR **M1** for 4, 5, 4 and 5, 4, 4 and 5, 5, 4 clearly identified on a tree or in a list
5(a)	15.6[0]	4	**B3** for $20\,900x = 326\,040$ or better or **M2** for $18\,500x + 2400(x - 2.5[0]) = 320\,040$ or **M1** for $18\,500x$ or $2400(x - 2.5[0])$
5(b)(i)	$(y+12)(y-7)$ final answer	2	**B1** for $(y+a)(y+b)$ where $ab = -84$ or $a+b=5$ or $y(y+12)-7(y+12)$ or $y(y-7)+12(y-7)$
5(b)(ii)	38 cao	3	**B2** for $y = 7$ or **M1** for $y(y+5) = 84$ oe
5(c)(i)	$168(m-0.75)+207m=100m(m-0.75)$ oe OR $207 = 100m - 168 - 75 + \frac{126}{m}$	M2	May be all over common denominator **M1** for $\frac{168}{m}$ or $\frac{207}{m-0.75}$ used
	at least one interim line leading to $50m^2 - 225m + 63 = 0$	A1	No errors or omissions

Question	Answer	Marks	Partial Marks
5(c)(ii)	$(10m-3)(5m-21)$ OR $m=\dfrac{-(-225)\pm\sqrt{(-225)^2-4(50)(63)}}{2(50)}$ oe OR $m=\dfrac{225}{100}\pm\sqrt{\left(\dfrac{225}{100}\right)^2-\dfrac{63}{50}}$ oe	**B2**	**M1** for $(10m+a)(5m+b)$ where $ab=63$ or $5a+10b=-225$ or $10m(5m-21)-3(5m-21)$ or $5m(10m-3)-21(10m-3)$ OR **M1** for $\sqrt{(-225)^2-4(50)(63)}$ or for $p=-(-225)$, $r=2(50)$ if in form $\dfrac{p+\sqrt{q}}{r}$ or $\dfrac{p-\sqrt{q}}{r}$ OR **M1** for $\left(m-\dfrac{225}{100}\right)^2$ oe
	4.2[0] cao	**B1**	
6(a)(i)	116.6 or 116.56 to 116.57	**4**	**M1** for $\sin[EAD]=\dfrac{6}{12}$ oe **M1** for $\tan[BAC]=\dfrac{6}{12}$ oe **B1** for [angle DAC] = 60
6(a)(ii)	13.4 or 13.41 to 13.42	**2**	**M1** for 12^2+6^2
6(a)(iii)	10.4 or 10.39…	**3**	**M2** for $\sqrt{12^2-6^2}$ or **M1** for $AE^2+6^2=12^2$
6(a)(iv)	130 or 129.5… to 129.6	**4**	**M1** for $0.5\times6\times theirAE$ oe **M1** for $0.5\times12\times12\times\sin 60$ oe **M1** for $0.5\times6\times12$ oe
6(b)(i)	3	**1**	
6(b)(ii)	51.3 or 51.30 to 51.34…	**4**	**M3** for tan = $\dfrac{8}{\sqrt{4^2+5^2}}$ or sin = $\dfrac{8}{\sqrt{4^2+5^2+8^2}}$ oe or **M2** for $\sqrt{4^2+5^2}$ or $\sqrt{4^2+5^2+8^2}$ or **M1** for angle ARB clearly indicated
7(a)	204 or 203.5 to 203.6… nfww	**4**	**M2** for $\pi\times1.5^2\times8\times60\times60$ or **M1** for $\pi\times1.5^2$ **M1** for dividing *their* volume by 1000 If 0 scored **SC1** for an answer figs 204 or figs 2035 to 2036 without working
7(b)(i)	$\pi\times6\times12+\pi\times6^2=108\pi$	**M2**	**M1** for $\pi\times6\times12$